

Individual Case Study

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Q1. Evaluate the consequences of offering a single ticket for the concert series either in place of or in addition to, offering tickets for each concert separately. Show your analysis using the numbers provided in the case for (1) selling individual concerts separately, (2) selling just the series (bundled concerts), and (3) selling both individual concerts as well as the series (bundle).

The orchestra has four equal-sized groups of potential customers, each with different willingness-to-pay for the two concerts. The four groups are Romantic, Neo-classical, Tchaikovsky lover, and Sophisticate.

(1) Selling the concerts separately

For the first concert, the willingness-to-pay amounts are \$40, \$20, \$45, and \$5. The orchestra can maximize revenue by charging **\$40**. At this price, the Romantic and Tchaikovsky lover groups would purchase tickets resulting in:

- $2(\$40) = \80

For the second concert, the willingness-to-pay amounts are \$20, \$40, \$5, and \$45. The orchestra can also maximize revenue by charging **\$40**. The Neo-classical and Sophisticate groups would purchase tickets:

- $2(\$40) = \80

Therefore, selling the concerts separately would generate:

- $\$80 + \$80 = \$160$

in revenue per four potential customers.

(2) Selling only the concert series

The combined willingness-to-pay for both concerts is:

Patron Type	Concert 1	Concert 2	Total Value
Romantic	\$40	\$20	\$60
Neo-classical	\$20	\$40	\$60
Tchaikovsky lover	\$45	\$5	\$50
Sophisticate	\$5	\$45	\$50

The orchestra could charge **\$50 for the series**, causing all four groups to purchase it. Revenue would therefore be:

- $4(\$50) = \200

This is \$40 more than selling the concerts separately.

(3) Selling both individual tickets and the series

The orchestra can charge **\$60 for the series** and **\$45 for individual concerts**.

At a \$60 series price, the Romantic and Neo-classical groups would purchase the series:

- $2(\$60) = \120

The Tchaikovsky lovers would instead purchase the first concert individually for \$45, while the Sophisticates would purchase the second individually for \$45.

Total revenue would therefore be:

- $\$120 + \$45 + \$45 = \210

Conclusion

Selling both individual tickets and the series generates the most revenue in this case. The results are:

- Individual concerts only: **\$160**
- Series only: **\$200**
- Individual tickets + series: **\$210**

Offering both options allows the orchestra to capture more of the different groups' willingness to pay.

Q2. Assume a potential market size of 20,000 consumers. Evaluate the consequences if the multiproduct company described in the case sells the two products only as a bundle. Show your analysis using the details provided in the case.

The company has a potential market of **20,000 consumers**. Each consumer has a willingness-to-pay for products A and B that is uniformly distributed from \$0 to \$100, and the willingness-to-pay for the two products is independent. The cost of providing either product is negligible.

Because the company sells only the bundle, let the bundle price be P .

For a bundle price P , the probability that a consumer purchases the bundle is:

- $1 - P^2 / 20,000$

Therefore:

- **Profit = $20,000P (1 - P^2 / 20,000)$**

Maximizing this expression gives:

- **$P = \sqrt{(20,000 / 3)} \approx \81.65**

At this price, approximately two-thirds of the market will purchase the bundle:

- $20,000(0.6667) \approx 13,333$ consumers.

The resulting profit is approximately:

- $13,333(\$81.65) \approx \$1,088,667$

Since the company's unit costs are negligible, revenue is essentially equal to profit.

Therefore, selling the two products only as a bundle at approximately \$81.65 would generate about \$1.089 million in profit from approximately 13,333 customers.

Q3. Assume the individual prices for product A and product B to be \$50. Evaluate the impact on profits if the company offers coupons, entitling a purchaser of product A to a certain amount (e.g., \$10) off the price of product B.

Assume both Product A and Product B are priced at \$50.

Without coupons, half of the 10,000 potential customers are willing to pay at least \$50 for each product.

For Product A:

- $10,000(0.50) = 5,000$ buyers.

For Product B:

- $10,000(0.50) = 5,000$ buyers.

Therefore, profit without coupons is:

- $5,000(\$50) + 5,000(\$50) = \$500,000$

Now suppose a purchaser of Product A receives a \$10 coupon for Product B. The coupon reduces the price of B from \$50 to \$40.

The 5,000 A buyers receive the coupon. Since the willingness-to-pay for B is uniformly distributed from \$0 to \$100, 60% of these consumers value B at \$40 or more:

- $5,000(0.60) = 3,000$

Therefore, 3,000 A buyers purchase B using the coupon:

- $3,000(\$40) = \$120,000$

There are also 2,500 consumers who do not purchase A but are willing to pay the full \$50 price for B:

- $5,000(0.50) = 2,500$

Their B purchases generate:

- $2,500(\$50) = \$125,000$

Revenue from A remains:

- $5,000(\$50) = \$250,000$

Thus, **total profit with the coupon** is:

- $\$250,000 + \$120,000 + \$125,000 = \$495,000$

Compared with the \$500,000 without coupons:

- $\$495,000 - \$500,000 = -\$5,000$

Therefore, offering the \$10 coupon reduces profit by \$5,000. Although the coupon increases the number of customers purchasing Product B, the \$10 discount reduces revenue enough that the company is worse off overall.

Q4. Assume the individual prices for product A and product B to be \$50. Would it be a good or bad idea to permit resale of the coupons amongst consumers? You can assume the transaction price of the coupon to be \$2. Provide a quantitative insight.

With both products priced at \$50, the company earns **\$500,000** when no coupons are offered.

If the \$10 coupons can be resold for **\$2**, the 5,000 consumers who purchase Product A initially receive coupons. Because a coupon can be sold for \$2, an A purchaser will use the coupon only if the value of using it is at least \$2 greater than selling it.

Therefore:

- **WTP for Product B - \$40 $\geq$ \$2**

or

- **WTP for Product B $\geq$ \$42**

Since willingness-to-pay is uniformly distributed from \$0 to \$100, 58% of the 5,000 Product A buyers will use their coupons:

- $5,000(0.58) = 2,900$

The **remaining coupons** are:

- $5,000 - 2,900 = 2,100$

These 2,100 coupons can be resold.

Among the 5,000 consumers who did not purchase Product A, 58% have a willingness-to-pay of at least \$42 for Product B:

- $5,000(0.58) = 2,900$

Because only 2,100 coupons are available for resale, there are more consumers who want the coupons than coupons available. Assuming the available coupons go to the consumers with the highest willingness-to-pay, the 2,100 resale buyers have a willingness-to-pay of at least \$58 for Product B. The remaining 800 consumers have willingness-to-pay between \$42 and \$58 and therefore do not purchase Product B at the regular \$50 price.

Product A revenue remains:

- $5,000(\$50) = \$250,000$

For **Product B**, 2,900 original coupon users and 2,100 resale coupon users pay \$40:

- $2,900(\$40) + 2,100(\$40) = \$200,000$

Therefore total profit is:

- $\$250,000 + \$200,000 = \$450,000$

Compared with the \$500,000 profit without coupon resale:

- $\$450,000 - \$500,000 = -\$50,000$

Therefore, permitting coupon resale would reduce the company's profit from **\$500,000 to \$450,000**, a decrease of **\$50,000**. The \$2 resale payment is transferred between customers rather than received by the company, while the resale of coupons allows additional consumers to purchase Product B at the discounted \$40 price.